Seth Balkrishan Memorial

BHILAI INSTITUTE OF TECHNOLOGY, DURG

(An Autonomous Institute affiliated to CSVTU, Bhilai)

SCHEME OF TEACHING AND EXAMINATION (Effective from 2020-2021 Batch)

B.Tech. (Information Technology) Eighth Semester

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
				L	T	P	Theory/Lab				
							ESE	CT	TA		
1.	Information Technology	Professional Elective-3	Refer Table-1	2	1	-	100	20	30	150	3
2.	Information Technology	Professional Elective-4	Refer Table-2	2	1	-	100	20	30	150	3
3.	Refer Table-3	Open Elective-3	Refer Table-3	3	-	-	100	20	30	150	3
4.	Information Technology	Machine Learning Lab	106891IT	-	-	2	25	-	25	50	1
5.	Information Technology	Hadoop Lab	106892IT	-	-	2	25	-	25	50	1
6.	Information Technology	Capstone Project Phase - 2	106893IT	-	-	18	300	-	150	450	9
Total				**7**	**2**	**22**	**650**	**60**	**290**	**1000**	**20**

Table 1 : Professional Elective -3

S. No.	Board of Study	Courses (Subject)	Course Code
1.	Information Technology	Machine Learning	106821IT
2.	Information Technology	Business Intelligence	106822IT
3.	Information Technology	Data Science	106823IT
4.	Information Technology	Block Chain Technology and Applications	106824IT
5.	Information Technology	Software Testing	106825IT

Table 2 : Professional Elective -4

S. No.	Board of Study	Courses (Subject)	Course Code
1.	Information Technology	Cyber Security	106826IT
2.	Information Technology	Big Data Analytics	106827IT
3.	Information Technology	Decision Support System	106828IT
4.	Information Technology	Deep Learning	106829IT
5.	Information Technology	5G Wireless and Mobile Technology	106830IT

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Table 3 : Open Elective -3					
S. No.	**Board of Study**	**Courses (Subjects)**	**Course Code**	**Eligible Branches**	**Link**
1.	Civil Engineering	Air Pollution and Control Measures	100841CE	Open to all branches	https://nptel.ac.in/courses/105107213
2.	Computer Science & Engineering	Machine Learning	100842CS	Open to all branches except CSE and IT	https://archive.nptel.ac.in/courses/106/105/106105152/
3.	Electrical & Electronics Engineering	Industrial Instrumentation	100843TE	Open to all branches	https://archive.nptel.ac.in/courses/108/105/108105064/
4.	Electrical Engineering	Biomedical Instrumentation	100844EE	Open to all branches	https://www.youtube.com/watch?v=i2mZylgP1Fk
5.	Electronics & Telecommunication Engineering	Reliability Engineering	100845ET	Open to all branches	https://onlinecourses.nptel.ac.in/noc23_ge20/preview
6.	Information Technology	Internet of Things	100846IT	Open to all branches except CSE and IT	https://nptel.ac.in/courses/106105166
7.	Mechanical Engineering	Industrial Work Ethics, Psychology and Sociology	100847ME	Open to all branches	https://onlinecourses.nptel.ac.in/noc19_hs35
8.	Management	Financial Management	100848MG	Open to all branches except students who have already opted the course in 6th Semester	https://archive.nptel.ac.in/courses/110/107/110107144/
9.	Computer Applications	Cloud Computing	100849CA	Open to all branches except CSE and IT	https://nptel.ac.in/courses/106105167

Note:

(a) Abbreviations used : L-Lecture, T- Tutorial, P- Practical, ESE- End Semester Exam, CT- Class Test, TA- Teacher's Assessment

(b) 1/4th of total strength of students subject to minimum of 20 students is required to offer an elective in the department in a particular academic session.

(c) Choice of elective course once made for an examination cannot be changed in future examinations.

(d) The duration of end semester examination of all theory papers will be of three hours.

SYLLABUS

B.TECH. (INFORMATION TECHNOLOGY)

EIGHTH SEMESTER

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Bhilai Institute of Technology, Durg

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SYLLABUS

B. Tech. (Information Technology) Eight Semester

MACHINE LEARNING					
Course Code	**106821IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To learn the different kinds of Learning techniques and get acquainted with the basics of machine learning methods and model validation methods and ways to measure their accuracy.	**On successful completion of the course, the student can**: **CO1.** Familiarize themselves with the machine learning goals and applications. **CO2.** Identify the various classification techniques and their performance evaluation methods **CO3.** Demonstrate various clustering methodologies and evaluation process **CO4.** Familiarize themselves with the different Machine learning curves **CO5.** Show the Neural network and Deep Learning concepts

UNIT I : Introduction to Machine Learning **[CO1]**

Introduction: Defining Learning systems, Goals and applications of Machine Learning, Aspects of developing a Learning system: training data, Concept representation, function approximation, Different form of learning **[7Hrs]**

UNIT II: Supervised Learning **[CO2]**

Entropy and Information Gain, K Nearest Neighbour- Challenges in KNN, Application of KNN, Decision trees

– ID3, Classificaiton and Regression Trees, Translating Decision tress into rules, Rule Based Classification, Over fitting, noisy data and pruning **[7Hrs]**

UNIT III : Unsupervised Learning **[CO3]**

Partioned Based Clustering, K – Means, K- Mediods, Hierarchical Clustering, Agglomerative, Divisive,

Distance Measure, DBSCAN, Density Based Clustering, Evaluation of Clustering method **[7Hrs]**

UNIT IV: Validations [**CO4**]

Validation Techniques, Need for Cross Validation, K-fold validation, Validation and Test Dataset, Confussion Matrix – Recall, Precision, Accuracy, F-Measure,
Learning Curves : ROC and AUC curve **[7Hrs]**

UNIT V: Introduction to Neural Network and Deep Learning [**CO5**]

Perceptron algorithm – Multilayer perceptrons back propagation nonlinear regression, Multicalss discrimination, Training procedures, Deep Learning overview and importance over machine Learning **[8Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Machine Learning	Tom M Mitchell	McGraw Hill
2	Introduction to Machine Learning	Ethem Alpaydin	MIT Press

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Introduction to Machine Learning	Nils Nilsson	MIT Press
2.	Readings in Machine Learning	Jude Shavil, Thomas G Dietterich	Morgan Kaufmann Publishers

Seth Balkrishan Memorial

Bhilai Institute of Technology, Durg

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SYLLABUS

B. Tech. (Information Technology) Eight Semester

BUSINESS INTELLIGENCE					
Course Code	**106822IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To become familiar with the ethics and basics of Business Intelligence and Decision Support Systems. To define mathematical models, data mining and data preparation	**On successful completion of the course, the student will be able to:** **CO1:** Explain the fundamental concepts, processes of business intelligence. **CO2**: Understand and use the technologies and tools that make up BI (e.g. Data warehousing, Data reporting and use of Online analytical processing (OLAP)). **CO3:** Implement BI techniques by using various tools and Create data visualization. **CO4:** Critically evaluate use of BI for supporting decision making in an organisation. **CO5** Apply various modeling techniques to solve real-world data analysis problems to various situations.

UNIT-I Introduction to Business Intelligence **[CO1]**

BI concept, BI architecture, BI in today's perspective, BI Process, Applications of BI like Financial analysis, statistical analysis, sales analysis, CRM, result pattern and ranking analysis, Balanced Scorecard, BI in Decision Modelling: Optimization, Decision making under uncertainty. Ethics and business intelligence. **[8Hrs]**

Unit II: Elements of Business Intelligence Solutions **[CO2]**

Reports & ad hoc queries; Analyse OLAP data; Dashboards & Scorecards development, Metadata Models; Automated tasks & events; Mobile & disconnected BI; Collaboration capabilities; Real time monitoring capabilities; Software development kit; Consume BI through portals, web applications, Desktop applications. **[7Hrs]**

UNIT-III Data Visualization and Dashboard Design, Performance Dashboard:[CO3]

Responsibilities of BI analysts by focusing on creating data visualizations and dashboards. Importance of data visualization, types of basic and composite charts. Measuring, Monitoring and management of Business, KPIs and dashboard, the types of dashboards, the common characteristics of Enterprise dashboard, design of enterprise dashboards, and the common pitfalls of dashboard design. **[7Hrs]**

Unit IV: Reporting authoring **[CO4]**

Building reports with relational vs Multidimensional data models ; Types of Reports – List, crosstabs, Statistics, Chart, map, financial etc; Data Grouping & Sorting, Filtering Reports, Adding Calculations to Reports, Conditional formatting, Adding Summary Lines to Reports. Drill up, drill- down, drill-through capabilities. Run or schedule report, different output forms – PDF, excel, csv, xml etc. **[7Hrs]**

UNIT-V Future of Business Intelligence: **[CO5]**

Emerging Technologies, Machine Learning, Predicting the Future with the help of Data Analysis, BI Search & Text Analytics – Advanced Visualization – Rich Report, and Future beyond Technology. **[7Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Fundamentals of Business Analytics	R. N. Prasad, Seema Acharya	Wiley-India
2.	Business Analytics for managers	Wolfgang Jank	Springer

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	"Decision Support and Business Intelligence Systems"	Efraim Turban, Ramesh Sharda, Dursun Delen,	Pearson
2.	Business Intelligence: The Savvy Manager"	David Loshin Morgan, Kaufman	Guide
3.	Business Intelligence Roadmap: The Complete Project Lifecycle of Decision Making",	Larissa T. Moss, S. Atre	Addison Wesley
4.	Business Intelligence: Data Mining and Optimization for Decision Making"	Carlo Vercellis	Wiley Publications
5.	The Kimball Group Reader: Relentlessly Practical Tools for Data Warehousing and Business Intelligence,	Ralph Kimball, Margy Ross	Wiley Publications

Seth Balkrishan Memorial

Bhilai Institute of Technology, Durg

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SYLLABUS

B. Tech. (Information Technology) Eight Semester

DATA SCIENCE					
Course Code	**106823IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
This course is intended to provide an introduction into the field of Data Science. As Data Science is a fast-growing interdisciplinary field, focusing on the analysis of data to extract knowledge and insight. This course will introduce students to the collection. Preparation, analysis, modeling and visualization of data, covering both conceptual and practical issues. work challenges.	**On successful completion of the course, the student will be able to:** **CO1**: Identify and describe the methods and techniques commonly used in data science. **CO2**: apply basic data cleaning techniques to prepare data for analysis. and Be able to identify the categorical and/or numerical data types in a given data set **CO3**: Demonstrate proficiency with the methods and techniques for obtaining, organizing, exploring, and analyzing data**.** **CO4**: Use appropriate tools and technology to collect, process, transform, summarize, and visualize data. **CO5:** Apply customizable tools for data analysis and visualization per the evaluation of characteristics of the data and the nature of the analysis.

Unit – I: Introduction **[CO1]**

Introduction to Data Science – Evolution of Data Science – Data Science Roles – Stages in a Data Science Project – Applications of Data Science in various fields – Data Security Issues. **[7Hrs]**

Unit – II: Data Representation **[CO2]**

Data Collection and Data Pre-Processing Data Collection Strategies – Data Pre-Processing Overview – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization. **[7Hrs]**

Unit – III: Data Analytics **[CO3]**

Exploratory Data Analytics Descriptive Statistics – Mean, Standard Deviation, Skewness and Kurtosis – Box Plots – Pivot Table – Heat Map – Correlation Statistics – ANOVA. **[7Hrs]**

Unit – IV: Model Development **[CO4]**

Model Development Simple and Multiple Regression – Model Evaluation using Visualization – Residual Plot – Distribution Plot – Polynomial Regression and Pipelines – Measures for In-sample Evaluation – Prediction and Decision Making. **[7Hrs]**

Unit – V: Model Evaluation **[CO5]**

Model Evaluation Generalization Error – Out-of-Sample Evaluation Metrics – Cross Validation – Overfitting – Under Fitting and Model Selection – Prediction by using Ridge Regression – Testing Multiple Parameters by using Grid Search. **[8Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Data Science and Big data Analytics	David Dietrich, Barry Heller, Beibei Yang	EMC

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Doing Data Science	Cathy O'Neil and Rachel Schutt	O'Reilly
2.	Handbook of Research on Cloud Infrastructures for Big Data Analytics	Raj, Pethuru	IGI Global
3.	Data Science from Scratch	Joel Grus	O'Reilly

Seth Balkrishan Memorial

Bhilai Institute of Technology, Durg

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SYLLABUS

B. Tech. (Information Technology) Eight Semester

BLOCK CHAIN TECHNOLOGY AND APPLICATIONS					
Course Code	**106824IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the students understand about the history, type and applications of block chain; also acquire knowledge about cryptography and consensus algorithms and finally deploy projects and design block chain based applications.	**On successful completion of the course, the student will be able to:** **CO1**: Contentedly discuss and describe the history, types and applications of Block chain. **CO2**: Experiment with cryptography and Consensus algorithms. **CO3**: Create and deploy projects using Web3j. **CO4**: Implement an ICO on Ethereal. **CO5**: Design block chain based application with Swarm and IPFS.

UNIT I: Introduction **[CO1]**

Distributed DBMS – Limitations of Distributed DBMS, Introduction to Blockchain – History, Definition, Distributed Ledger, Blockchain Categories – Public, Private, Consortium, Block chain Network and Nodes, Peer-to-Peer Network, Mining Mechanism, Generic elements of Blockchain, Features of Block chain, and Types of Blockchain. **[7Hrs]**

UNIT II: Architecture of Blockchain **[CO2]**

Operation of Bitcoin Blockchain, Blockchain Architecture – Block, Hash, Distributer P2P, Structure of Blockchain- Consensus mechanism: Proof of Work (PoW), Proof of Stake (PoS), Byzantine Fault Tolerance(BFT), Proof of Authority (PoA) and Proof of Elapsed Time (PoET). **[8Hrs]**

UNIT III: Blockchain Based Future System **[CO3]**

Project presentation- Futures smart contract: Blockchain oracles- Web3j: Setting up the Web3J- Installing web3j- Wallet creation, Java client: The wrapper generator- Initializing web3j- Setting up Ethereum accounts- Deploying the contract. **[7Hrs]**

UNIT IV: Blochchains in Business **[CO4]**

Public versus private and permissioned versus permission less blockchains- Privacy and anonymity in Ethereum; The Ethereum Enterprise Alliance- Blockchainas-a-Service- Initial Coin Offering (ICO): Project setup for ICO implementation- Token contracts- Token sale contracts- Contract security and testing the code CO-4. **[7Hrs]**

UNIT V:Distributed Storage IPFS AND SWARM **[CO5]**

Ethereum Virtual Machine- Swarm and IPFS: Installing IPFS, Hosting our frontend: Serving your frontend using IFPS, Serving your frontend using Swarm. **[7Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Mastering Blockchain: Distributed Ledger Technology, decentralization, and smart contracts explained	Imran Bashir	Packt Publishing Ltd
2.	Blockchain By Example: A developer's guide to creating decentralized applications using Bitcoin, Ethereum, and Hyperledger	Bellaj Badr, Richard Horrocks, Xun (Brian) Wu	Packt Publishing Ltd

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Mastering Bitcoin: Unlocking Digital Cryptocurrencies	Andreas M. Antonopoulos	O'Reilly Media Inc
2.	Bitcoin and Cryptocurrency Technologies: A comprehensive Introduction	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder,	Princeton University Press

SOFTWARE TESTING					
Course Code	**106825IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To study software testing objectives, process, criteria, strategies, and methods and various software testing issues and solutions in software unit, integration, regression, and system testing.	**On successful completion of the course, the student will be:** **CO1:** to design and conduct a software test process for a software testing project. **CO2:** to analyze concepts of software testing theory. **CO3:** to identify various control flow technique for testing problems. **CO4:** to solve problems by designing and selecting software test models, criteria, strategies, and methods. **CO5:** to use software testing methods and modern software testing tools for their testing projects.

UNIT I: **[CO1]**
Quality Revolution, Software Quality, Role of Testing, Objectives of Testing, Concept of Complete Testing, Central Issue of Testing, Sources of Information for Test Case selection, Test Planning and Design, Monitoring and Measuring Test Execution, Test Tools and Automation, Test Team Organization and Management. **[7Hrs]**

UNIT II: **[CO2]**
Basic Concepts of Testing Theory, Theory of Goodenough and Gerhart, Theory of Weyuker and Ostrand, Theory of Gourlay, Adequacy of Testing, Limitations of Testing, Static Unit Testing, Defect Prevention, Dynamic Unit Testing, Debugging. **[9Hrs]**

UNIT III: **[CO3]**
Outline of Control Flow Testing, Control Flow Graph, Paths in Control Flow Graphs, Path Selection Criteria, Data Flow Testing criteria, Comparison of Data Flow and Test Selection Criteria, Domain Error, Testing of Domain Errors. **[8Hrs]**

UNIT IV: **[CO4]**
System Test design, Test design Factors, Requirement Identification, Test Objective Identification, Structure of a System Test Plan, Assumptions, Test Approach, Test Suite Structure, Types Of Acceptance Testing **[6 Hrs]**

UNIT V: **[CO5]**
Five Views of Software Quality, Quality Control, Quality assurance, Cost of quality, Software Quality Assurance, SQA Plan, ISO 9000, Capability Maturity Model, McCall s Quality Factors. **[6 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Software Testing and Quality Assurance	Kshirasagar Naik,	John Wiley & Sons.
2.	Effective Methods for Software Testing	William Perry	John Wiley & Sons.

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Testing Computer Software	Cem Kaner and Jack Falk	Wiley
2	Software Testing	Ron Patton	SAMS Publications

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SYLLABUS

B. Tech. (Information Technology) Eight Semester

CYBER SECURITY					
Course Code	**106826IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To prepare students with the technical knowledge and skills needed to protect and defend computer systems and networks. To develop graduates that can plan, implement, and monitor cyber security mechanisms to help ensure the protection of information technology assets.	**On successful completion of the course, the student will be able to:** **CO1:** Explain with the latest cyber security news and trends and make sure you are implementing adequate cyber security measures**.** **CO2:** Examine threats and risks within context of the cyber security. **CO3:** Interpret and forensically investigate security incident. **CO4:** Apply policies and procedures to manage Privacy issues**.** **CO5:** Describe key terms and concepts in cyber law, intellectual property and cybercrimes, trademarks and domain theft.

Unit – 1: Cyber Security Fundamentals **[CO1]**

Cyber security: Introduction, Internet society, Cyber security objectives, Cyber security roles, Cyber security Common attack types and vectors, Issues and challenges of cyber security.

Cyber Crimes and Criminals: Definition of cyber-crime, Types of Cyber Crime, Classification of Cyber Criminals, Tools used in Cyber Crime: Challenges, Strategies. **[8Hrs]**

Unit – 2: Cyber attacker Techniques and Motivations **[CO2]**

Anti-forensics: Use of proxies, use of tunneling techniques. Fraud techniques: Phishing and malicious mobile code, Rogue antivirus, Click fraud. Threat Infrastructure: Botnets, Fast Flux and advanced fast flux. **[7Hrs]**

Unit – 3: Web Application Vulnerabilities **[CO3]**

Understanding vulnerabilities in traditional client server application and web applications, cookie based attacks, SQL injection, cross domain attack. SSL vulnerabilities and testing Cross-site request forgery, Social Engineering, War Xing, Brute force and dictionary attacks, DNS Amplification Attacks. **[6Hrs]**

Unit – 4: Information Technology Act 2000 **[CO4]**

Introduction – Need for Cyber Law - Evolution of the IT Act, Genesis and Necessity, Salient features of the IT Act, 2000, various authorities under IT Act and their powers, Penalties & Offences, Amendments and Limitations of IT Act., Electronic Governance, Legal Recognition of Electronic Records, Legal Recognition of Digital Signature, Certifying Authorities, Cyber Crime and Offenses. **[8Hrs]**

Unit – 5: Cyber Law and Related Legislation **[CO5]**

Intellectual Property Rights: Patent Law, Trademark Law, Copyright, Copyright in the Digital Medium, Copyright in Computer Programmes, Software Copyright, Electronic Data Base and its Protection, IT Act and Civil Procedure Code, IT Act and Criminal Procedural Code, Law Relating To Employees And Internet, Alternative Dispute Resolution, Online Dispute Resolution (ODR). **[7Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Cyber Forensics	Dejey, Dr.Murugan	Oxford University Press
2.	Cyber Security Essentials	James Graham et al	CRC Press
3.	Cyber Laws: Intellectual property & E Commerce Security	Kumar K.	Dominant Publisher

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives	Sumit Belapure and Nina Godbole	Wiley India Pvt. Ltd.
2.	Ethical Decision making and IT: An Introduction with Cases	Earnest A. Kallman, J.P Grillo	McGraw Hill Publication.
3.	Fundamentals of Network Security	E. Maiwald	McGraw Hill
4.	Cyber Law and IT Protection	Harish Chander	PHI Publication
5.	Computer Security: Principles and Practice	William Stallings and Lawrie Brown	Prentice Hall.

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SYLLABUS

B. Tech. (Information Technology) Eight Semester

BIG DATA ANALYTICS					
Course Code	**106827IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The Objective of the course is to explore the fundamental concepts of Big Data Analytics and Visualization Techniques, learn to use various techniques for Mining Data Streams, to understand Big Data Analytics and Hadoop concepts, to understand the Hadoop Framework and various Big Data enabling Technologies, to understand Open Source Database Concepts.	On successful completion of the course, the student will be able to: **CO1:** Identify the concepts of Big Data Analytics and visualization techniques. **CO2:** Familiarized themselves the various techniques used for mining data stream. **CO3:** Familiarized themselves the Hadoop Concepts. **CO4:** Outline the Hadoop frameworks and big data enabling Technologies. **CO5:** Demonstrate the concepts of Open Source databases such as NOSQL, HBase etc.

UNIT-I: INTRODUCTION TO BIG DATA: **[CO1]**

Introduction to Big Data, Characteristics of Big Data(5 V's Of Big Data),Sources of Big Data, Challenges of Conventional Systems, Analysis vs Analytics, Types of Data Analytics, Analysisvs Reporting ,Visual data analysis. **[8hrs]**

UNIT-II: MINING DATA STREAMS: **[CO2]**

Introduction To Streams Concepts, Stream Data Model and Architecture, Sampling Data in a Stream, Filtering Streams, Counting Distinct Elements in a Stream, Estimating Moments **[6hrs]**

UNIT-III: HADOOP: **[CO3]**

Components of Hadoop, The Hadoop Distributed File System (HDFS), Map Reduce, YARN, COMMON, Hadoop Cluster, How Map Reduce Works, Anatomy of Map Reduce, Different types of Schedulers in Hadoop **[7hrs]**

UNIT-IV: HADOOP FRAMEWORKS: **[CO4]**

Applications on Big Data Using Pig, Hive, Zookeeper, Hadoop frameworks- Cassandra, Apache Mahout ,Tez, Hbase, Avro, Chukwa, Spark, Ambari. **[7hrs]**

UNIT-V: NOSQL: **[CO5]**

Limitations of SQL Databases, Introduction to NoSQL databases, types of NoSQL Databases, Document Databases, Key Valued Databases, Column based Databases and Graph Databases. Issues of availability and consistency, CAP Theorem, Case studies of MongoDB, Neo4j, HBase, Cassandra **[8hrs]**

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Text Books:

S.No.	Title	Author(s)	Publisher
1.	Intelligent Data Analysis: An Introduction	Michael R. Berthold, David J. Hand	Springer
2.	Hadoop: The Definitive Guide	Tom White	O'Reilly Media, Inc.
3.	Mining of Massive Datasets	Jure Leskovec, Anand Rajaraman, Jeffrey David Ullman	Cambridge University Press

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Big Data in Practice	Bernard Marr	Willy publications
2.	Too Big to Ignore: The Business Case for Big Data	Phil Simon	Willy publications

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SYLLABUS

B. Tech. (Information Technology) Eight Semester

DECISION SUPPORT SYSTEM					
Course Code	**106828IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
Objectives to include this subject in Information Technology discipline is: • To review and clarify the fundamental terms, concepts and theories associated with Decision Support Systems, computerized decision aids, expert systems, group support systems and executive information systems.	**On successful completion of the course, the student can**: **CO1.** Recognize the relationship between business information needs and decision making. **CO2.** Recognize the general nature and range of decision support systems **CO3.** State the issues related to the Analyse, design, development and implement a DSS. **CO4.** Select appropriate modeling techniques. **CO5.** Justify or identify the requirements of group and intelligent decision support system.

UNIT I: OVERVIEW OF DIFFERENT TYPES OF DECISION-MAKING [CO1]

Strategic, tactical and operational. Consideration of organizational structures. Mapping of databases, MIS, EIS, KBS, expert systems OR modeling systems and simulation, decision analytic systems onto activities within an organization. Extension to other 'non organizational' areas of decision making. Relationship with knowledge management systems. **[7Hrs]**

UNIT II: HUMAN COGNTION IN DECISION [CO2]

Studies of human cognition in relation to decision making and the assimilation of information. Cultural issues. Implications for design of decision-making support. Communication issues. **[9Hrs]**

UNIT III: Analytics [CO3]

Normative, descriptive and prescriptive analysis: requisite modeling. Contrast with recognition primed decision tools Knowledge management initiatives, Approaches to Knowledge management **[8 Hrs]**

UNIT IV: MODELING TOOLS [CO4]

Database, MIS, EIS, KBS, Belief nets, data mining. OR modeling tools: simulation and optimization. History, design, implementation: benefits and pitfalls. Risk assessment. Decision analysis and strategic decision support. **[6Hrs]**

UNIT V: GROUP AND INTELLIGENT DECISION [CO5]

Group decision support systems and decision conferencing. Intelligent decision support systems: tools and applications. Cutting-edge decision support technologies. History, design, implementation: benefits and pitfalls. Deliberative e-democracy and e-participation **[6Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Decision Sciences: an integration Perspective	P.R. Kleindorfer, H.C. Kunreuther , P.J.H. Schoemaker	Cambridge University Press
2.	Decision support Systems in the 21st Century	G.M. Marakas	Prentice Hall

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Decision support Systems and Intelligent Systems	E. Turban and J.E. Aronson	Prentice Hall
2.	Decision Support Systems	V.S.Janakiraman and K.Sarukesi	PHI
3.	Decision Support and Data Warehouse Systems	Efrem G. Mallach	TMH

BHILAI INSTITUTE OF TECHNOLOGY, DURG

(An Autonomous Institute affiliated to CSVTU, Bhilai)

SCHEME OF TEACHING AND EXAMINATION (Effective from 2020-2021 Batch)

B.Tech. (Information Technology) Eighth Semester

DEEP LEARNING					
Course Code	**106829IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
Objectives to include this subject in Information Technology discipline is: •Introduce the student to the latest algorithms which have made everything from Self driving cars to Google's Alpha Go possible. Deep Learning is at the core of modern-day Artificial Intelligence and Machine Learning applications. Student will be introduced to the theoretical background required to fully understand the ongoing research in the field.	**On successful completion of the course, the student will be able to**: **CO1.** Explain machine learning basics and back propagation model. **CO2.** Regularize the deep models. **CO3.** Describe the basic concepts and Apply optimization techniques for training deep models. **CO4.** Apply convolution networks for computer vision problems. **CO5.** Apply and Use Deep Recurrent Architecture.

UNIT I: Feed forward Deep Networks **[CO1]**

Review of Machine Learning Basics, Vanilla MLP, Estimating conditional Statistics, Flow Graphs and Back propagation, Universal Approximation Properties and Depth, Feature representation, Piecewise Linear Hidden UNITs. **[7Hrs]**

UNIT II: Regularization of Deep Models **[CO2]**

Regularization from Bayesian Perspective, Parameter Norm Penalty, Regularization as Constrained Optimization, Under-Constrained Problems, Classical Regularization as Noise Robustness, Dropout, Multi-Task training, Adversarial Training. **[9Hrs]**

UNIT III: Optimization for Training Deep Models **[CO3]**

Optimization for Model Training, Challenges in Optimization, Basic Algorithms, Adaptive learning rates, Second order methods, Natural gradient methods, Global Optimization. **[8Hrs]**

UNITIV: Convolutional Networks **[CO4]**

The Convolution Neural Network, Filters, Convolutions Pooling, Convolution models, Efficient Convolution Algorithms, Random or Unsupervised features, Keraas, Transfer Training Applications in Computer Vision. **[6Hrs]**

UNIT V: Recurrent and Recursive Nets **[CO5]**

Unfolding Flow graphs and parameter sharing, Recurrent Neural Networks, Input output sequences, Training RNNs: Backpropagation Through Time, Bidirectional RNNs, Deep Recurrent Architecture, Auto- Regressive Networks. **[6Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Deep Learning	Ian Good fellow, Yoshua Bengio and Aaron Courville.	

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Artificial Intelligence for Humans: Deep Learning and Neural Networks,	Jeff Heaton	Heaton Research, Incorporated.
2	Deep Learning with Tensor Flow,	Giancarlo Zaccone	Packt Publishing

5G WIRELESS AND MOBILE TECHNOLOGY					
Course Code	**106830IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The aim of this course is to let the students differentiate between 2G, 3G, 4G and 5G. Whereas, 5G aims to provide extremely low delay services, great service in crowd, enhanced mobile broadband (virtual reality being made real), ultra-reliable, ubiquitous QoS, and highly energy efficient networks	**On successful completion of the course, the student can**: **CO1.** Understand and explain the continuous evolution of different cellular generations. **CO2.** Analyze the various use cases and learn basics of 5G. **CO3.** Draw and explain 5G architecture, its components and functional criteria. **CO4.** Understand various possible 5G networks in future. **CO5.** Analyze the recent scenarios of 5G deployment and possible challenges in future

UNIT I: Introduction To Wireless Communication Systems **[CO1]**

Introduction to wireless communication systems- Evolution of mobile communication system- 2G, 3G,4G, UMTS, LTE. Wireless transmission, Frequencies, Signals, Antennas, Signal propagation, Multiplexing, Modulation, Spread spectrum, Cellular Systems: Cellular concept and frequency reuse, Frequency Management and Channel Assignment, Types of handoff and their characteristics. **[7Hrs]**

UNIT II: Cellular Concepts And System Design FUNDAMENTALS **[CO2]**

GSM: Architecture, Location tracking and call setup, Mobility management, Handover, Security, subscriber and service data management, Evolution of high speed data techniques in existing scenarios, Future trends for enhanced data rate. **[9Hrs]**

UNIT III: Mobile Network And Transport Layer **[CO3]**

IP and Mobile IP Network Layers, Packet Delivery and Handover Management, Location Management, Registration, Tunneling and Encapsulation, Route Optimization, DHCP. Traditional TCP, Indirect TCP, Snooping TCP, Mobile TCP, Fast retransmit / Fast recovery, Transmission / Timeout freezing, Selective retransmission, Transaction Oriented TCP. **[8Hrs]**

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UNIT IV: Mobile Ad hoc Networks (MANETs) : **[CO4]**

Introduction, Applications & Challenges of a MANET, Routing, Classification of Routing Algorithms, Algorithms such as DSR, AODV, DSDV, etc., Mobile Agents, Service Discovery **[6Hrs]**

UNIT V: Current state and Challenges ahead **[CO5]**

5G penetration in developed countries; deployment challenges in low-middle income countries stronger backhaul requirements, dynamic spectrum access and usage of unlicensed spectrum. Contrasting radio resource requirements; large cell usage: LMLC; possible solutions for connectivity in rural areas (BharatNet, TVWS, Long-range WiFi, FSO); non-terrestrial fronthaul/backhaul solutions:LEOs.HAP/UAV. **[6Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	4G, LTE-Advanced Pro and The Road to 5G	Erik Dahlman, Stefan Parkvall, Johan Skold	Elsevier Science
2.	5G NR: Architecture, Technology, Implementation, and Operation of 3GPP New Radio Standards Hardcover	Sassan Ahmadi Dr	Academic Press

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Fundamentals of 5G Mobile Networks	Jonathan Rodriguez	Wiley
2.	5G Mobile and Wireless Communications Technology	Afif Osseiran, Jose F. Monserrat, Patrick Marsch	Cambridge University Press

Online References:

S.No.	Website Name	URL
1.	NPTEL	https://nptel.ac.in/courses/108/105/108105134/
2.	Udemy	https://www.udemy.com/course/5g-mobile-networksmodern-wireless-communication-technology

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Bhilai Institute of Technology, Durg

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SYLLABUS

B. Tech. (Civil Engineering) Eighth Semester

AIR POLLUTION AND CONTROL MEASURES					
(Open Elective Course : Open to all branches)					
Course Code	**100841CE**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
To know about air pollution sources, classification, and role of meteorological parameters in atmospheric stability conditions and plume behaviors. To learn about types of air pollution and their source, effect and control techniques. To find the flue gases concentration using Gaussian plume modelling and understand functioning of different sampling devices. To understand the deterioration mechanism of air pollutants and major disaster episodes. To learn about air quality parameters and their global effects.	**On successful completion of the course, the student will be able to:** **CO1:** Discuss about the basics of air pollution and meteorological parameter in atmospheric conditions. **CO2:** Discuss about the sources of air pollution their effects and control techniques. **CO3:** Describe Air pollution monitoring. **CO4:** Discuss mechanism of air pollution and its effects. **CO5:** Demonstrate the understanding of air quality criteria and emission standards, air pollution act and global effects of air pollution.

UNIT – I: Air Pollution **CO1**

Problem, Definitions, Classification of pollutants, characteristics, and sources.
Atmospheric Conditions: Meteorology Parameters, Atmospheric stability class condition, plume behavior, wind roses. **[6 Hrs.]**

UNIT – II: Air Pollutants **CO2**

SOX sources, ambient concentrations, SOX control techniques, effects of SOX on human, animal health, plants and on materials. NOX sources, ambient concentrations, control techniques, effects of NOX on human health, animal health, plants and on materials.
Particulate Matter sources and size distribution, effects of PM on human, animal health, plants and on materials, control techniques **[8 Hrs.]**

UNIT – III: Air pollution Monitoring **CO3**

Gaussian plume models, assumptions and limitations of GPM, problem on modelling, Measurement of stack gases, Sampling methods, Difficulties in sampling, sampling of SPM, stack sampling techniques. **[8 Hrs.]**

UNIT – IV: Mechanism of Air pollution and its effects **CO4**

Deterioration Mechanisms of air pollutants and effect of air pollution on art treasures in India, Major air pollution disaster episodes,
Special diseases caused by air pollution, symptoms of chronic air pollution, effects due to Topographical on air pollution, **[7 Hrs.]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

UNIT – V: Air Quality Emission Standards **CO5**

Air quality criteria and emission standards, US and Indian standards, air pollution act, constitution, power, and functions of the boards.

Global and regional Environmental Issues: Global effects of air pollution – Greenhouse effect, Global warming, Acid rains, Ozone layer depletion **[7 Hrs.]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Environmental Pollution Control Engineering	C. S Rao	New Age International
2.	Environmental Engineering	Peavy & Rowe	Tata McGraw Hill
3.	Environmental Science and Engineering	Henry and Heinke	Pearson Education

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Air Pollution	Henry C. Perkins	McGraw Hill
2.	Introduction to Environmental Science	Y. Anjaneyulu	Tata McGraw Hill
3.	Waste Water Engineering	Metcalf Eddy	Tata McGraw Hill

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SYLLABUS

B. Tech. (Computer Science & Engineering) Eighth Semester

MACHINE LEARNING					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code :	**100842CS**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
Course objective is to make student learn the basics of Machine Learning. The students would learn about the various framework for building machine learning systems. They will know what is supervised and unsupervised learning. Also they will gain knowledge about Artificial Neural Networks.	**On successful completion of the course, the student will be able to:** **CO1:** Explain the basic concepts of machine learning **CO2:** Demonstrate the use of various frameworks of machine learning systems. **CO3**: Apply supervised machine learning and its various methods of implementation. **CO4:** Apply unsupervised machine learning and its various methods of implementation. **CO5:** Explain the use of ANN in solving real world problems.

UNIT I: **(CO1)**

Introduction to Machine Learning: History and Evolution, Artificial Intelligence Evolution, Machine Learning Categories: Supervised Learning, Unsupervised Learning and Reinforcement Learning. **[7 HRS]**

UNIT II: **(CO2)**

Frameworks for Building Machine Learning Systems: SEMMA Model (Sample, Explore, Modify, Model, Assess), Machine Learning Perspective of Data: Nominal scale, Ordinal scale, Interval scale and Ratio scale, Dealing with Missing Data, Handling Categorical Data, Normalizing Data. **[8 HRS]**

UNIT III: **(CO3)**

Supervised Learning: Linear Regression, Polynomial Regression, Logistic Regression, Decision Trees, Random Forest Algorithm. **[7 HRS]**

UNIT IV: **(CO4)**

Unsupervised Learning: Introduction, K Means Clustering, Principal Component Analysis (PCA). Bias and Variance, K-Fold Cross-Validation, Bagging, Boosting. **[7 HRS]**

UNIT V: **(CO5)**

Artificial Neural Network (ANN): Introduction, Perceptron, Multilayer Perceptrons (Feed forward Neural Network), Restricted Boltzman Machines (RBM). **[7 HRS]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Mastering Machine Learning with Python in Six Steps	Manohar Swamynathan	Apress
2.	A Concise Introduction to Machine Learning	A. C. Faul	CRC Press
3.	Applied Machine Learning	David Forsyth	Spinger

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Applied Machine Learning with Python	Andrea Giussani	BUP
2.	Practical Machine Learning Cookbook	Atul Tripathi	Packt

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SYLLABUS

B. Tech. (Electrical and Electronics Engineering) Eighth Semester

INDUSTRIAL INSTRUMENTATION					
(Open Elective Course : Open to all branches)					
Course Code	100843TE	L = 3	T = 0	P = 0	Credits = 3
Examination Scheme	ESE	CT	TA	Total	ESE Duration
	100	20	30	150	3 Hours
	Minimum number of class tests to be conducted = 02			Minimum assignments = 02	

Course Objectives	Course Outcomes
The course provides theoretical knowledge of various methods used for the measurement of force, speed, viscosity, humidity, flow, temperature & pressure and also explains the importance of instrumentation in process industry.	**On successful completion of the course, the student will be able to:** **CO1:** Explain static characteristics of industrial instrumentation & importance of instrumentation in process industry. **CO2:** Describe various methods to measure pressure. **CO3:** Describe various methods for measurement of temperature. **CO4:** Explain flow measurement methods & flow meters. **CO5:** Discuss various apparatus used for measurement of humidity, viscosity, force & speed.

UNIT – I **(CO1)**

Introduction: Mechanical Vs Electrical instruments; Measurement & Instrument; measurement methods; Static terms & Characteristics: Range & Span, Accuracy Error & Correction, Calibration, Sensitivity, Threshold & Resolution, Precision & Repeatability; Dynamic terms & Characteristics: Speed of Response, Fidelity, Overshoot, Dead time & Zone. **[8Hrs]**

UNIT – II **(CO2)**

Pressure Measurement: Introduction, Terminology used in pressure measurement, pressure units & measuring instruments, Manometers, Mechanical displacement type gauges, Mechanical Gauges: Bourdon tube & Diaphragms, Bellows. **[7 Hrs]**

UNIT – III **(CO3)**

Temperature Measurement: Introduction, Temperature Scales, Classification of temperature measuring devices, liquid-in-glass thermometer, pressure-gas thermometer, liquid filled systems, gas filled systems, Thermocouple, Resistance Thermometer & Thermister**.** **[7Hrs]**

UNIT – IV **(CO4)**

Flow Measurement: Introduction, Nature of flow, Flow measuring Techniques, Flow measurement methods, Venturi flow meter, Flow Nozzles, Orifice flow meter, Electromagnetic flow meter, Hot wire anemometer**.** **[7Hrs]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

UNIT – V **(CO5)**

Miscellaneous Measurement (Humidity, Viscosity, Force & Speed):

Humidity Measurement Methods: Dry bulb temperature, wet bulb temperature, Resistive hygrometers, Capacitive hygrometers, Dew point temperature.

Viscosity Measurement: Saybolt viscometer, Falling sphere viscometer, Falling piston viscometer.

Force Measurement: Pneumatic load Cell, Hydraulic load cell, Strain Gauge load cell.

Speed Measurement: AC & DC Tachogenerators**.** **[7Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Mechanical Measurements and Instrumentation and Control	A.K. Sawhney and Puneet Sawhney	Dhanpat Rai
2.	Mechanical Measurements & Control	Dr. D. S. Kumar	Metropolitan Book
3.	Measurement systems Application and Design	E. O. Doebelin and D. N. Manik.	McGraw-Hill Education

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Instrumentation Engineers Handbook (Measurement)	B. G. Liptak	CRC Press
2.	Principles of Industrial Instrumentation	D. Patranabis	McGraw-Hill Education
3.	Industrial Instrumentation	D. P. Eckman	Wiley Eastern Limited
4.	Industrial Instrumentation and Control	S. K. Singh	Tata Mc-Graw-Hill

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SYLLABUS

B. Tech. (Electrical Engineering) Eighth Semester

BIOMEDICAL INSTRUMENTATION					
(Open Elective Course : Open to all branches)					
Course Code	**100844EE**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
• To make electrical engineers aware about human physiology Basics. • To have an idea about the electrical parameters related to human body and to perform their measurements.	**On successful completion of the course, the student will be able to:** **CO1.** Explain an acquaintance of the physiology of the heart, lung, blood circulation and circulation respiration. **CO2.** Describe about various sensing and measurement devices of electrical origin. **CO3.** Illustrate the latest ideas on devices of non-electrical devices. **CO4.** Describes the important and modern methods of imaging techniques. **CO5.** Describes latest knowledge of medical assistance / techniques and therapeutic equipment.

UNIT – I **(CO1)**

Human Physiology and Basics: Brief introduction to human physiology, Basic components of bio-medical instruments, bioelectric signals, action potentials, Bio-electrodes. **[7Hrs]**

UNIT – II **(CO2)**

Transducers: Biomedical Transducers: displacement, velocity, force, acceleration, flow, temperature, potential, dissolved ions and gases. **[7Hrs]**

UNIT – III **(CO3)**

Electro-Physiological Measurements: Analysis of EEG, ECG, EMG, EOG, & Bio-Potential Amplifiers for ECG, EMG, EEG, etc. **[6Hrs]**

UNIT – IV **(CO4)**

Electrical Parameter Measurements: Cardiovascular measurement-blood pressure, blood flow, stroke volume, Impedance Plethysmography, Cardiac output, heart sound etc. Instrumentation for respiratory & nervous systems. **[6Hrs]**

UNIT – V **(CO5)**

Monitoring, Assisting, Therapeutic Equipment and Safety Patient care & monitoring system, Remote monitoring through telephone, Internet, Satellite link, Safety aspects associated with Biomedical Instrumentation. Recent advances in Bio-Medical Instrumentation, Microprocessor based systems, Laser & optical Fiber systems. **[6Hrs]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Biomedical Instrumentation and Measurements	Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer	Prentice-Hall
2.	Handbook of Biomedical Instrumentation	R. S. Khandpur,	McGraw Hill.
3.	NPTEL Link: https://www.youtube.com/watch?v=i2mZylgP1Fk		

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Biomedical Instrumentation	M. Arumugam	Anuradha Agencies.
2.	Introduction to Biomedical Engineering	Domach	Pearson Education.

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SYLLABUS

B. Tech. (Electronics and Telecommunication Engineering) Eighth Semester

Reliability Engineering					
(Open Elective Course : Open to all branches)					
Course Code	100845ET	L = 3	T = 0	P = 0	Credits = 3
Examination Scheme	ESE	CT	TA	Total	ESE Duration
	100	20	30	150	3 Hours
	Minimum number of class tests to be conducted = 02			Minimum assignments = 02	

Course Objectives	Course Outcomes
The primary objective of the course is to demonstrate the approaches and techniques to assess and improve process and/or product quality and reliability.	**On successful completion of the course, the student will be able to:** **CO1:** Explain major concepts of reliability and fault models of engineering systems. **CO2**: Analyze statistical experiments leading to reliability modelling. **CO3**: Analyze the causes of failure and use preventive maintenance techniques. **CO4**: Apply Reliability theory for assessment of reliability in engineering design. **CO5:** Analyze different methods for improving reliability.

UNIT – I **(CO1)**

Basics Of Reliability: Mathematics of Reliability, Reliability function, Models of failure. Failure data Analysis, System Reliability. **[8Hrs]**

UNIT – II **(CO2)**

Reliability Models And Systems: Basic Reliability Models, Covariate Models, Hazard Rate Functions including Exponential, Weibull, Normal and Lognormal, System Reliability including redundant, standby and load sharing systems. **[7Hrs]**

UNIT – III **(CO3)**

Reliability And Failure: Failure Mode, Effect and Criticality Analysis, Fault Tree Analysis, Reliability andMaintainabilityDesignMethodsBasedonAvailabilityandLifeCycleCosts, Preventive Maintenance. **.** **[7Hrs]**

UNIT – IV **(CO4)**

Failure Preparedness: Spares Provisioning Models, Renewal and Minimal Repair Models, Treatment of Censored Data, Reliability Growth Testing, Probability Tests And Curve Fitting, Maintaining Likelihood Estimation and Goodness of Fitness Tests, Series Configuration, Parallel Configuration, R-Out-Of-N Structure. **[7Hrs]**

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UNIT – V **(CO5)**

Improvement And Checks: Reliability improvement. Redundancy, Reliability Allocation, Reliability Testing**.** **[7Hrs]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

Text Books:

S. No.	Title	Author(s)	Publisher
1.	An Introduction to Reliability and Maintainability Engineering	Ebeling	Tata McGraw Hill
2.	Probabilistic Reliability - An Engineering Approach	M.L. Shooman	McGraw Hill

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Engineering Design Reliability Handbook	Boca Raton	CRC Press
2.	Reliability Engineering	E. Balagurusamy	McGraw Hill

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SYLLABUS

B.Tech. (Eighth Semester)

INTERNET OF THINGS					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100846IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
Course objective is to make the students to learn the basics, characteristics, evolution, challenges and application areas of IoT, get an idea of protocol organization, create an overall view of various communication aspects of IoT, understand the cloud and internet environment, make able to develop IoT system, understand the various case studies.	**On successful completion of the course, the student will be able to:** **CO1:** Identify the components of IoT **CO2:** Analyze various protocols of IoT. **CO3:** Design sensors based system using Arduino board. **CO4:** learn basic wireless networking with ESP8266 module and interface Raspberry Pi **CO5:** Identify and use various Cloud Platforms for IOT.

UNIT I **(CO1)**

Introduction to IoT: Definition, Origin, Characteristics, IoT Market Share, Evolution of Connected Devices, Sensing, Actuation, Basics of IoT Networking, IoT Categories, IoT Challenges. **[6 Hrs]**

UNIT II **(CO2)**

IoT Protocols Organization: Communication Protocol: IEEE 802.15.4, Zigbee, 6LowPAN, RFID, NFC, Bluetooth.

Data Protocols: MQTT, CoAP, XMPP, AMQP; M2M vs IoT. **[8 Hrs]**

UNIT III **(CO3)**

Arduino Simulation Environment: Arduino Uno Architecture, Setup the IDE, Writing Arduino Software, Arduino Libraries, Basics of Embedded C programming for Arduino, Interfacing LED, push button and buzzer with Arduino, Interfacing Arduino with LCD.

Sensor & Actuators with Arduino: Overview of Sensors working, Analog and Digital Sensors, Interfacing of Temperature, Humidity, Motion, Light and Gas Sensor with Arduino, Interfacing of Actuators with Arduino, Interfacing of Relay Switch and Servo Motor with Arduino. **[8 Hrs]**

UNIT IV **(CO4)**

Basic Networking with ESP8266 WiFi module: Basics of Wireless Networking, Introduction to ESP8266 Wi-Fi Module, Various Wi-Fi library, Web server- introduction, installation, configuration, Posting sensor(s) data to web server.

Raspberry Pi – Installation and interfaces (Serial, SPI, I2C) **[8 Hrs]**

UNIT V **(CO5)**

Cloud Platforms for IOT: Virtualization concepts and Cloud Architecture, Cloud computing, benefits, Cloud services -- SaaS, PaaS, IaaS, Cloud providers & offerings, Study of IOT Cloud platforms, ThingSpeak API & Arduino Cloud, Interfacing ESP8266 with Web services. **[6 Hrs]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Internet of Things (A Hands-On-Approach)	Arshdeep Bahga, Vijay Madisetti	VPT
2.	The Internet of Things in the Cloud: A Middleware Perspective	Honbo Zhou	CRC Press
3.	The Internet of Things – Key Applications and Protocols.	Olivier Hersent, David Boswarthick, Omar Elloumi	Wiley
4.	Learning Internet of Things	Peter Waher	Packt Publishing

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Architecting the Internet of Things	Dieter Uckelmann, Mark Harrison	Springer
2.	Fundamentals of Wireless Sensor Networks: Theory and Practice	Waltenegus Dargie, Christian Poellabauer	Wiley
3.	The Internet of Things: A Survey	Luigi Atzori, Antonio Lera, Giacomo Morabito	Elsevier
4.	[Online] https://nptel.ac.in/courses/106/105/106105166/		

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SYLLABUS

B.Tech. (Mechanical Engineering) Eighth Semester

INDUSTRIAL WORK ETHICS , PSYCHOLOGY AND SOCIOLOGY					
(Open Elective Course : Open to all branches)					
Course Code	**100847ME**	**L =3**	**T =0**	**P=0**	**Credits=3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
This course aims to provide the students a conceptual and analytical orientation to the society and its dynamics so that as individuals they may be able to understand the realities and identify themselves as part of the larger system. To outline various psychological, sociological and ethical aspects in the context of industrial and social environments	**On successful completion of the course, the student will be able to:** **CO1:** Discuss about professional work ethics and Social responsibility. **CO2:** Demonstrate the understanding of the basic responsibilities and rights. **CO3:** Demonstrate the understanding of the global work standards and ethical practices. **CO4:** Discuss key ideas concerning industrial psychology. **CO5:** Explain the conventional and contemporary theories of work towards the sociological analyses of work and industry.

UNIT–I: PROFESSIONAL WORK ETHICS **CO1**

Ethics in Workplace - Formulation of Ethics - Managerial Ethics - Managing Ethical Behavior - Codes of Ethics - Encouraging Ethical Behavior - Ethical Leadership - Ethical Decision making. Corporate Social Responsibility (CSR) - Intellectual Property Rights (IPR)- Meaning Laws relating to Intellectual Property Rights (IPRs) **[7Hrs]**

UNIT–II: RESPONSIBILITIES & RIGHTS **CO2**

Collegiality and loyalty, Managing conflict, Respect for authority, Collective bargaining, confidentiality, Role of confidentiality in moral integrity, Conflicts of interest, Occupational crime, Professional rights, Employee right, IPR Discrimination. **[7Hrs]**

UNIT– III: GLOBAL ETHICS ISSUES **CO3**

Multinational Corporations- Environmental Ethics- Business Ethics- Computer Ethics -Role in Technological Development- Engineers as Managers- Consulting Engineers- Engineers as Expert witnesses and advisors-Moral leadership. **[7Hrs]**

UNIT–IV: INDUSTRIAL PSYCHOLOGY **CO4**

Introduction of Psychology as a science- area of application- study of individual-Individual differences-study of behavior-stimulus response behavior-heredity and environment- Motivational theory by McClelland-Affiliation motivation, Power motivation-Maslow's Hierarchy of Needs-Herzberg's two factor model-Alderf's ERG Model. **[7Hrs]**

		July 2023	1.00	Applicable for AY 2023-24Onwards
Chairman (AC)	Chairman (BoS)	Date of Release	Version	

UNIT–V: SOCIOLOGY **CO5**

Industrial Sociology: Nature, Scope and Importance of Industrial Sociology. Social Relations in Industry, Social Organization in Industry-Bureaucracy, Scientific Management and Human Relations. Modern Sociological Theories: Scientific Management Theory of F.W. Taylor, Elton Mayo's Human Relations Approach, Harry Braverman's Labour and Monopoly Capital, Maslow theory of Industrial Management, Development of Industrial Structure: Informal and Formal Industry, Marx's theory of alienation, Division of Labour, Alienation, Labour Unrest and Class-Struggle. **[7Hrs]**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Work Ethics	Narendra Modi Ji	Prabhat Prakashan
2.	A text book on Professional Ethics & Human Values	R.S. Naagarasan	New Age International

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Professional Ethics and Human Values	M. Govindarajan	PHI Learning Pvt. Ltd.
2.	Handbook of Industrial, Work and Organizational Psychology	Neil Anderson, Deniz S Ones, Handan Kepir Sinangil & Chockalingam Viswesvaran	SAGE Publications,
3.	Sociology of Work and Professional Ethics	Miljenko Antić	Faculty of Civil Engineering, Zagreb
4.	The Sociology of Work	Steven P. Vallas, William Finlay, and Amy S. Wharton	Oxford University Press
5.	Fundamentals of Industrial sociology	Gisbert Pascal	Tata McGraw Hill,
6.	Industrial Psychology	VikramBisen & Priya	New Age International
7.	Business Ethics	Murthy C.S.V	Himalaya Publishing House

Seth Balkrishan Memorial

Bhilai Institute of Technology, Durg

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SYLLABUS

B.Tech. (Eighth Semester)

FINANCIAL MANAGEMENT					
(Open Elective Course : Open to all branches except students who have already opted the course in 6th Semester)					
Course Code	**100848MG**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
1. The objective of this course is to understand various concepts related to financial management. 2. To study in detail various tools and techniques in the area of finance. 3. To develop the analytical skills that would facilitate financial decision making.	**On successful completion of the course, the student will be able to:** **CO1:** Classify funding sources and demonstrate knowledge of value of money over time. **CO2:** Examine complexities associated with financing decision. **CO3:** Demonstrate understanding of the concept of budgeting and evaluate proposals. **CO4**: Select and apply techniques in management of working capital. **CO5:** Interpret the profit distribution decisions.

UNIT–I Introduction to Financial Management **CO1**

Financial Management: Nature and Objectives, Profit maximization v/s Wealth maximization, Finance Function, Time value of money-Discounting and Compounding Techniques, Long term and Short-term sources of Finance, Introduction to Capital Market. **[8HRS]**

UNIT–II Financing Decision **CO2**

Cost of Capital: Weighted Average Cost of Capital

Capital Structure: Factors, Approaches and Theories

Leverage: Operating and Financial Leverage: Impact, Trading on Equity **[7HRS]**

UNIT–III Investment Decision **CO3**

Budget: Concept and Types, Budgetary Control, Capital Budgeting, Zero based Budgeting.

. **[7HRS]**

UNIT–IV Working Capital Decision **CO4**

Management of Working Capital: Concept, Need, Factors and Estimation of Working Capital, Inventory and Receivables Management, Management of Cash **[7HRS]**

UNIT–V Dividend Decision **CO5**

Dividend Policy: Types, Factors Influencing Dividend Policy and Dividend Models. **[7HRS]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Financial Management	Eugene F Brigham	Cengage Learning
2.	Financial Management	MY Khan,P K Jain	McGraw Hill

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Financial Decision Making: Concepts, Problems and Cases	John J. Hampton	PHI
2.	Financial Management and Policy	V. K. Bhalla	Anmol Publications
3.	Financial Management	Tulsian & Tulsian	S Chand
4.	Essentials of Financial Management	I. M. Pandey	Vikas Publishing House
5.	Financial Management	P. Chandra	McGraw Hill

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SYLLABUS

B.Tech. (Eighth Semester)

CLOUD COMPUTING					
(Open Elective Course : Open to all branches except CSE & IT)					
Course Code	**100849CA**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The Objective of this course is to make student know about Cloud computing emergence, its offering, benefits, characteristics and Types of cloud. Also to make them aware about cloud securities and management policy for identity and security also to managing service provider and data center about virtualization concepts.	**On successful completion of the course, the student will be able to:** **CO1:** Identify the basic services offered by cloud computing and its benefits. **CO2:** Familiarize themselves securities offer by the cloud. **CO3:** Demonstrate others services provided by cloud. **CO4:** Show the services provided by Managed services. **CO5:** Outline the concepts of virtualization.

UNIT I: : **Introduction to Cloud Computing :** **(CO1)**

Introduction to Cloud Computing, The Emergence of Cloud Computing, Cloud-Based Service Offerings, Benefits of using a Cloud Model, Key Characteristics of Cloud Computing, Understanding Public & Private cloud environments, The Evolution of Cloud Computing – Hardware & Internet Software Evolution. **[8 Hrs]**

UNIT II: Cloud Security and Management: **(CO2)**

Cloud Security Challenges, Software-as-a-Service, Security Management People, Security Governance, Security Portfolio Management, Security Architecture Design, Identity Access Management (IAM), Data Security. **[7 Hrs]**

UNIT III: Cloud Services: **(CO3)**

Cloud as: Communication-as-a-Service (CAAS), Infrastructure-as-a-Service (IAAS), Monitoring-as-a Service (MAAS), Platform-as-a-Service (PAAS), Software-as-a-Service (SAAS). **[7 Hrs]**

UNIT IV: Managed Service Provider and Data center: **(CO4)**

The MSP Model, Evolution from the MSP Model to Cloud Computing and Software-as-a-Service, Service oriented Architecture, Cloud Data Center **[7 Hrs]**

UNIT V: Virtualization Concepts: **(CO5)**

Virtualization concepts: virtualization benefits, Hardware virtualization, Software Virtualization, Memory Virtualization, Storage Virtualization, Data Virtualization, Network Virtualization, Virtualization Security Recommendations. **[7 Hrs]**

		July 2023	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2023-24 Onwards

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Cloud Computing: A Practical Approach.	Toby Velte, Anthony Vote and Robert Elsenpeter	McGraw Hill
2.	Cloud Application Architectures: Building Applications and Infrastructures in the Cloud	George Reese	O'Reilly Media

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance	Tim Matherm, Subra Kumaraswamy and Shahed Latif	O'Reilly Media

MACHINE LEARNING LAB					
Course Code	**106891IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The objective of the course is aimed to gain basic concepts and techniques of Machine Learning along with real time problem solving technique using various Machine Learning techniques	**On successful completion of the course, the student will be able to:** **CO1.** Select real-world applications that needs machine learning based solutions. **CO2.** Implement and apply machine learning algorithms. **CO3.** Select appropriate algorithms for solving a particular group of real-world problems. **CO4.** Recognize the characteristics of machine learning techniques that are useful to solve real world problems **CO5.** Demonstrate the Neural network and Deep Learning concepts

Experiment to be performed

1. Exercises to solve the real-world problems using Linear Regression machine learning methods. [**CO1]**
2. To apply Logistic Regression Method for solving real world problem. [**CO1]**
3. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample. [**CO2]**
4. To study about Multi-Class Classification for Deep machine learning. [**CO2]**
5. Implement a Neural Network for a classification Problem. Observe the accuracy of the network by varying the number of layers, number of neurons in each layer, activation function, number of epochs, learning rate etc. [**CO3]**
6. To apply Support Vector Machines K Method for solving real world problem. [**CO3]**
7. To study about Means Clustering & PCA for Deep machine learning. [**CO4]**
8. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a.CSV file. Compute the accuracy of the classifier, considering few test data sets. [**CO4]**
9. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem. [**CO5]**
10. Use some research data set from ML dataset repositories to train a machine learning model that generates useful insights for the domain. [**CO5]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, And Techniques to Build Intelligent Systems	A Geron	O Reilly
2	Learning TensorFlow: A Guide to Building Deep Learning Systems	Tom Hope, Yehezkel S Resheff	O Reilly
3	Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2,	Sebastian Raschka	Packt

HADOOP LAB					
Course Code	**106892IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
This course will introduce the basics required to develop map reduce programs; derive business benefit from unstructured data. This course will also give an overview of the architectural concepts of Hadoop and introducing map reduce paradigm. Another objective of this course is to introduce programming tools PIG & HIVE in Hadoop ecosystem.	**On successful completion of the course, the student will be able to:** **CO1:** Identify Big Data framework like Hadoop. **CO2:** List the components of Hadoop and Hadoop Eco-System. **CO3:** Implement the file management tasks in Hadoop. **CO4:** Program applications using tool likes Hive, pig, No SQL, MongoDB for Big data Applications **CO5:** Construct scalable Algorithms to solve Data Intensive Problems using Map Reduce Paradigm.

List of Experiments

(At least Ten experiments are to be performed by each student)

1.	Installation of Single Node Hadoop Cluster on Ubuntu	**(CO1)**
2.	**Study of HADOOP eco system.**	**(CO2)**
3.	**Basic HDFS commands**	**(CO2)**
4.	Hadoop File System navigation and manipulation using commands.	**(CO3)**
5.	Implement the following file management tasks in Hadoop: i. Adding files and Directories ii. Retrieving files iii. Deleting files	**(CO3)**
6.	Implementing simple algorithm in Map-Reduce : Matrix Multiplication, Aggregates, Joins, sorting, searching etc.	**(CO5)**
7.	Hadoop Programming: Word Count Map Reduce Program.	**(CO5)**
8.	Hive sell Writing basic Hive queries: Hive DDL, Hive DML.	**(CO4)**
9.	Using Hive to perform CRUD operations: Database, Tables, Views, Functions, Indexes	**(CO4)**
10.	Use Hives to create alter and drop database, tables, views, functions, and indexes.	**(CO4)**
11.	Participating pig commands from grunt shell.	**(CO4)**
12.	Writing pig scripts and running them.	**(CO3)**
13.	Processing different datasets using pig.	**(CO3)**

Text Books:

S. No.	Title	Author(s)	Publisher
1.	Big Java	Cay Horstmann	Wiley John Wiley & Sons
2.	Hadoop: The Definitive Guide	Tom White	O'reilly

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Hadoop MapReduce Cookbook	Srinath Perera, Thilina Gunarathne	O'reilly
2.	Hadoop for Dummies	Dirk deRoos, Paul C.Zikopoulos, Roman B.Melnyk,Bruce Brown, Rafael Coss	Wiley John Wiley & Sons
3.	Hadoop in Practice	Alex Holmes	MANNING Publication

CAPSTONE PROJECT PHASE - 2					
Course Code	**106893IT**	**L = 0**	**T = 0**	**P = 18**	**Credits = 9**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	300	**-**	**150**	**450**	**-**

Course Objectives	Course Outcomes
The Course objective is to provide an opportunity for students to understand real-life practices and implementation of theoretical Engineering knowledge in practical environment. To provide an opportunity for students to acquire interpersonal skills and ability for team work. To obtain knowledge of how to make optimal decisions to resolve technical challenges.	On successful completion of the course, the student will be able to: **CO1**: Apply engineering knowledge in real life projects by integrating theoretical concepts with real-time situations. **CO2:** Acquire practical skills and Communication skills while working on projects. **CO3:** Obtain knowledge of how to make optimal decisions to resolve technical challenges. **CO4:** Write technical documents and give oral presentations related to the project work completed. **CO5:** Use of Advanced Tools for developing better projects.

- The Capstone Project Phase-2 will consist of the work on the topic selected for the project.
- The topics should be a real time application or research or innovations in engineering field.
- The project must be done in a group not exceeding four students.
- The candidate is required to complete the implementation of the project work which was started in Phase-1 of last semester.
- Team Work: The assessments of the team work should be done by two internal examiners, one of which will be the guide and the other will be HOD or project coordinator from the department.
- Guidelines for the term work and assessment shall be as described in the Phase-1 of the same program. The review committee may be constituted by the Head of the Department.
- The progress of the project is evaluated based on a minimum of two reviews. Practical Examination: Practical Examination will consist of a presentation along with actual demonstration of the project.
- The candidate will submit project report to project guide and head of the department.
- The internal evaluation will be made based on the product, the report and a viva-voce examination, conducted internally by a committee appointed by Head of the Department comprising HoD or a Project coordinator for that program.